

ENRICHMENT

17. In this question you may assume that the trajectory of a projectile is a parabola, and that its directrix is independent of α , the angle of projection. Suppose it is found that there is only one angle at which a projectile may be fired from point A to pass through point P .
- (a) Use geometry to show that the focus of the trajectory is on the line AP .
- (b) Now prove the result algebraically. You may need the results of Question 14.

6H Resisted Projectile Motion

In what follows, it is assumed that a particle of mass m is projected from the origin with initial velocity vector $u_0\mathbf{i} + v_0\mathbf{j}$ and is under the influence of gravity in a medium with resistance. Typically the resistance is proportional to $|v|$ or $|v|^2$. The main focus of this section is the case where the resistance is proportional to $|v|^2$. There is no analytic solution to this problem, but solutions can be found in various cases where it is appropriate to approximate the resistance with another function. As will be seen, the mathematics is quite complicated but may provide a rewarding investigation for students able to take on the challenge.

Vectors and Calculus: It is often convenient to use vectors for projectile motion. However, rates are involved so the derivative is required, and the derivative of a vector has not yet been defined. Here is an outline of what is required.

Suppose that a particle has position vector $p(t) = x(t)\mathbf{i} + y(t)\mathbf{j}$ at time t . The velocity of the particle is then defined to be

$$v = \lim_{h \rightarrow 0} \frac{1}{h} (p(t+h) - p(t)),$$

which is analogous with the definition for a simple function of time. Expanding this definition to the component form gives

$$v = \lim_{h \rightarrow 0} \left(\frac{x(t+h) - x(t)}{h} \mathbf{i} + \frac{y(t+h) - y(t)}{h} \mathbf{j} \right),$$

thus $v = \dot{x}(t)\mathbf{i} + \dot{y}(t)\mathbf{j}$.

Likewise the acceleration vector is defined to be

$$a = \lim_{h \rightarrow 0} \frac{1}{h} (v(t+h) - v(t)) \\ = \lim_{h \rightarrow 0} \left(\frac{\dot{x}(t+h) - \dot{x}(t)}{h} \mathbf{i} + \frac{\dot{y}(t+h) - \dot{y}(t)}{h} \mathbf{j} \right),$$

thus $a = \ddot{x}(t)\mathbf{i} + \ddot{y}(t)\mathbf{j}$.

As can be seen from each case above, it is simply a matter of differentiating the components to obtain the required result. Though it will not be needed here, the same is true in three dimensions, and the proof is left as an exercise.

Many interesting results arise from this definition of differentiation. The most obvious is that the velocity vector is tangent to the path of the particle at each point. This is easily shown in two dimensions as follows. The gradient at any point on the path is

$$\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx} \quad (\text{by the chain rule}) \\ = \dot{y}(t) \div \dot{x}(t) \quad (\text{treating the derivative like a fraction.})$$

For these reasons, the above approximations and solutions should be discarded. The results are unreliable as they were obtained from inconsistent approximations and an error, replacing $\sqrt{v^2}$ with v instead of $|v|$. In some cases, plotting the equations above gives a solution curve which is not far from that obtained by the numerical approximation as outlined in Exercise 6E, but this is more a matter of good fortune rather than good design.

Low-angle Trajectory: When the angle of projection is small, the initial vertical component of velocity will be small. Further, the vertical velocity will approach zero near the turning point. Thus it is clear that $u \gg v$ for the entire journey. Consequently $(u^2 + v^2)^{\frac{1}{2}} \approx |u|$ and $|u| = u$, since the horizontal component of velocity is always positive by observation. Thus equations (1) and (2) above may be approximated with

$$\dot{u} = -ku^2 \quad (3)$$

$$\text{and } \dot{v} = -g - kuv. \quad (4)$$

These equations are relatively easy to solve. Also, because the approximation is valid at every point on the curve, there should be good agreement between the solutions of (3) and (4) and the solutions of (1) and (2). Solving (3) first, which is variable separable,

$$u = \frac{u_0}{1 + ku_0 t}$$

$$\text{and } x = \frac{1}{k} \log(1 + ku_0 t).$$

Equation (4) then becomes

$$\dot{v} = -g - \frac{ku_0 v}{1 + ku_0 t}.$$

This equation can be solved by rearranging it as follows.

$$(1 + ku_0 t)\dot{v} + ku_0 v = -g(1 + ku_0 t).$$

Notice that the left hand side is the derivative of a product, so

$$\frac{d}{dt}((1 + ku_0 t)v) = -g(1 + ku_0 t)$$

which, after integration, applying the initial conditions and a few lines of algebra,

$$\text{gives } v = -\frac{g(1 + ku_0 t)}{2ku_0} + \frac{v_0 + \frac{g}{2ku_0}}{1 + ku_0 t}$$

$$\text{and } y = \frac{g(1 - (1 + ku_0 t)^2)}{(2ku_0)^2} + \left(v_0 + \frac{g}{2ku_0}\right) \frac{1}{ku_0} \log(1 + ku_0 t).$$

High-angle Trajectory: When the angle of projection is nearly a right angle, the horizontal component of velocity is tiny and so $v \gg u$ for the majority of the journey. Consequently $(u^2 + v^2)^{\frac{1}{2}} \approx |v|$ and so equations (1) and (2) above may be approximated with

$$\dot{u} = -ku|v| \quad (5)$$

$$\text{and } \dot{v} = -g - kv|v|. \quad (6)$$

As with vertical projections, the upward and downward journeys are considered separately. For the upward journey, $|v| = v$ so that

$$\dot{u} = -kuv$$

$$\text{and } \dot{v} = -g - kv^2.$$

The solution for the vertical component was given above and is

$$v = \sqrt{\frac{g}{k}} \tan(\theta - \sqrt{gk}t)$$

$$\text{and } y = \frac{1}{k} \log \left(\frac{\cos(\theta - \sqrt{gk}t)}{\cos \theta} \right)$$

where $\theta = \tan^{-1} \left(\sqrt{\frac{k}{g}} v_0 \right)$ and v_0 is the vertical component of the initial velocity. Notice that $v = 0$ when $t = \frac{\theta}{\sqrt{gk}}$, so this solution is valid for $0 \leq t \leq \frac{\theta}{\sqrt{gk}}$.

The equation for $\dot{u}$ then becomes

$$\dot{u} = -u\sqrt{gk} \tan(\theta - \sqrt{gk}t)$$

which is variable separable. A few lines of working yield

$$u = \frac{u_0 \cos \theta}{\cos(\theta - \sqrt{gk}t)}$$

Now put $\phi = (\theta - \sqrt{gk}t)$ to get

$$\begin{aligned} x &= -\frac{u_0 \cos \theta}{\sqrt{gk}} \int_{\theta}^{(\theta - \sqrt{gk}t)} \sec \phi \, d\phi \\ &= \left[-\frac{u_0 \cos \theta}{\sqrt{gk}} \log \left(\frac{1 + \tan \frac{\phi}{2}}{1 - \tan \frac{\phi}{2}} \right) \right]_{\theta}^{(\theta - \sqrt{gk}t)} \quad (\text{see Exercise 4G Q16}) \\ &= \left[\frac{u_0 \cos \theta}{\sqrt{gk}} \log \left(\frac{1 - \tan \frac{\phi}{2}}{1 + \tan \frac{\phi}{2}} \right) \right]_{\theta}^{(\theta - \sqrt{gk}t)} \\ &= \left[\frac{u_0 \cos \theta}{\sqrt{gk}} \log \tan \left(\frac{\pi}{4} - \frac{\phi}{2} \right) \right]_{\theta}^{(\theta - \sqrt{gk}t)} \\ &= \frac{u_0 \cos \theta}{\sqrt{gk}} \log \left(\frac{\tan(\frac{\pi}{4} - \frac{\theta}{2} + \frac{1}{2}\sqrt{gk}t)}{\tan(\frac{\pi}{4} - \frac{\theta}{2})} \right). \end{aligned}$$

For the downward journey, $|v| = -v$ so equations (5) and (6) become

$$\dot{u} = kuv$$

$$\text{and } \dot{v} = -g + kv^2.$$

Notice that the terminal velocity is $V_T = -\sqrt{\frac{g}{k}}$.

The solution of the vertical component is:

$$v = -\sqrt{\frac{g}{k}} \times \frac{e^{\tau\sqrt{gk}} - e^{-\tau\sqrt{gk}}}{e^{\tau\sqrt{gk}} + e^{-\tau\sqrt{gk}}}$$

$$\text{and } y = \frac{1}{k} \log \left(\frac{2 \sec \theta}{e^{\tau\sqrt{gk}} + e^{-\tau\sqrt{gk}}} \right)$$

$$\text{or } y = \frac{1}{k} \log \left(\frac{2\sqrt{g + kv_0^2}}{\sqrt{g}(e^{\tau\sqrt{gk}} + e^{-\tau\sqrt{gk}})} \right)$$

where $\tau = t - \frac{\theta}{\sqrt{gk}}$, that is, the time after starting the downward journey. Now put the result for v into the horizontal component to get

$$\dot{u} = -ku \sqrt{\frac{g}{k}} \times \frac{e^{\tau\sqrt{gk}} - e^{-\tau\sqrt{gk}}}{e^{\tau\sqrt{gk}} + e^{-\tau\sqrt{gk}}}$$

which is variable separable yielding

$$u = \frac{2u_0 \cos \theta}{e^{\tau\sqrt{gk}} + e^{-\tau\sqrt{gk}}}$$

although, for the purposes of integration, it is better to write

$$u = 2u_0 \cos \theta \times \frac{e^{-\tau\sqrt{gk}}}{1 + e^{-2\tau\sqrt{gk}}}$$

$$\text{Thus } x = \frac{2u_0 \cos \theta}{\sqrt{gk}} \left[\frac{\pi}{4} - \frac{1}{2} \log \tan \left(\frac{\pi}{4} - \frac{\theta}{2} \right) - \tan^{-1} \left(e^{-\tau\sqrt{gk}} \right) \right].$$

Note that near the top of the trajectory $v \rightarrow 0$ and thus $u \gg v$ in that vicinity. Consequently the approximation $(u^2 + v^2)^{\frac{1}{2}} \approx |v|$ is not valid near the turning point. Thus it is expected that the high-angle trajectory approximation will be good for most of the journey, but poor near the turning point.

For those familiar with them, the integration for the downward journey would be much simpler were the hyperbolic trigonometric functions available, since then

$$v = -\sqrt{\frac{g}{k}} \tanh(\tau\sqrt{gk}),$$

$$y = \frac{1}{k} \log \left(\sec \theta \operatorname{sech}(\tau\sqrt{gk}) \right)$$

$$\text{and } u = u_0 \cos \theta \operatorname{sech}(\tau\sqrt{gk}).$$

Mid-angle Trajectory: When the angle of projection is about 45° , both components of the velocity will be roughly equal for a significant portion of the trajectory. It seems reasonable then to approximate $(u^2 + v^2)^{\frac{1}{2}}$ with either $\sqrt{2}u$ or $\sqrt{2}|v|$. That is, equations (1) and (2) will be approximated with

$$\dot{u} = -k\sqrt{2}u^2 \quad (7)$$

$$\text{and } \dot{v} = -g - k\sqrt{2}|v|v, \quad (8)$$

Notice that these equations are similar to the erroneous approximation that began this discussion. The two crucial differences are the factor of $\sqrt{2}$ and the presence of the absolute value function $|v|$, with the need to divide the trajectory into upward and downward journeys.

Further notice that equation (7) may be obtained by replacing k with $k\sqrt{2}$ in equation (3). Thus equation (7) has solution

$$u = \frac{u_0}{1 + k\sqrt{2}u_0t}$$

$$\text{and } x = \frac{1}{k\sqrt{2}} \log(1 + k\sqrt{2}u_0t).$$

Likewise, equation (8) may be obtained by replacing k with $k\sqrt{2}$ in equation (6). Thus equation (8) has solution for the upward journey

$$v = \sqrt{\frac{g}{k\sqrt{2}}} \tan(\theta - \sqrt{gk\sqrt{2}}t)$$

$$\text{and } y = \frac{1}{k\sqrt{2}} \log \left(\frac{\cos(\theta - \sqrt{gk\sqrt{2}}t)}{\cos \theta} \right)$$

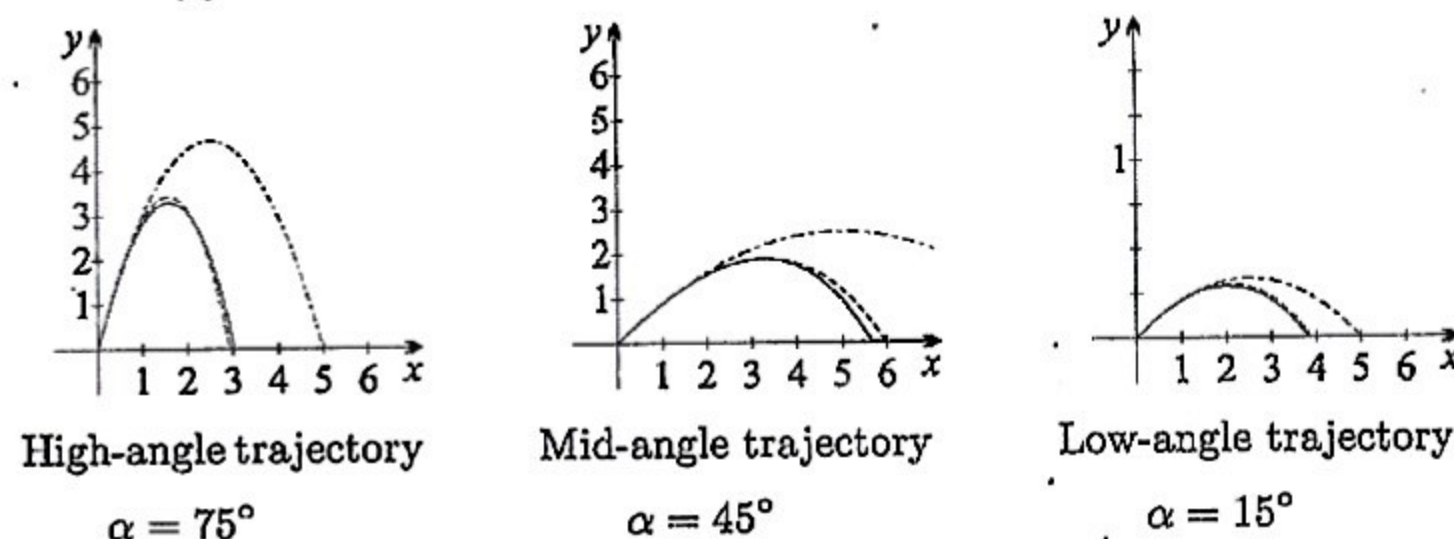
where $\theta = \tan^{-1} \left(\sqrt{\frac{k\sqrt{2}}{g}} v_0 \right)$ and v_0 is the vertical component of the initial velocity. Notice that the vertical velocity is zero when $t = \frac{\theta}{\sqrt{gk\sqrt{2}}}$, so this solution is valid for $0 \leq t \leq \frac{\theta}{\sqrt{gk\sqrt{2}}}$. The solution for the downward journey is

$$v = -\sqrt{\frac{g}{k\sqrt{2}}} \times \frac{e^{\tau\sqrt{gk\sqrt{2}}} - e^{-\tau\sqrt{gk\sqrt{2}}}}{e^{\tau\sqrt{gk\sqrt{2}}} + e^{-\tau\sqrt{gk\sqrt{2}}}}$$

$$\text{and } y = \frac{1}{k\sqrt{2}} \log \left(\frac{2\sqrt{g + k\sqrt{2}v_0^2}}{\sqrt{g}(e^{\tau\sqrt{gk\sqrt{2}}} + e^{-\tau\sqrt{gk\sqrt{2}}})} \right)$$

where $\tau = t - \frac{\theta}{\sqrt{gk\sqrt{2}}}$, that is, the time after starting the downward journey.

Graphs of the Approximations:



In each case the initial velocity is $|u_0| = 10$, acceleration due to gravity is $g = 10$, the coefficient of resistance is $k = 0.1$, the terminal velocity is $V_T = 10$ and the angle of projection is α . The solid line is the appropriate approximation. The dashed line is the solution to the full equations (1) and (2) obtained numerically using a vector form of the fourth order Runge-Kutta method, an advanced and extremely reliable approximation method which is studied in some university courses. The trajectory with no resistance is shown with a dash-dot line. As predicted in the discussion above, the high-angle approximation is less accurate near the turning point and consequently also on the downward journey, but is close to the actual trajectory overall. The mid-angle also breaks down near the turning point as foretold, but again is a good approximation overall.

As was expected, the low-angle approximation gives excellent results. Notice the change in vertical scale in the third graph. Even with this vertical exaggeration, the difference between the full equation solution and the approximation is barely noticeable.

The Exercise Questions: There is a limited number of questions that can be asked on resisted projectile motion. A few examples for the case where resistance is proportional to the velocity have been included in Sections 6E and 6F.

The questions in the following exercise involve the approximations considered above for when resistance is proportional to the square of the velocity. There are two questions on the low-angle trajectory and one each on the other two approximations. They are difficult, mostly due to the complicated integration required. Consequently there are no Foundation level questions. The solutions are lengthy, and so the style of the questions may not be suitable for an examination or similar assessment. Each question, however, would make an excellent basis for a project or research assignment.

Exercise 6H

DEVELOPMENT

1. [Low-angle Trajectory] A particle is projected from the origin with speed 10 m/s at an angle of inclination of 15° . Resistance is proportional to the square of the speed at each point in the trajectory and the constant of proportionality is $\frac{1}{10}m$ where m is the mass of the particle. Assume that the acceleration due to gravity is 10 m/s^2 . Also assume that the speed may be approximated by $(u^2 + v^2)^{\frac{1}{2}} \doteq u$. The equations of motion are thus

$$\dot{u} = -\frac{1}{10}u^2 \quad (3)$$

$$\text{and } \dot{v} = -10 - \frac{1}{10}uv. \quad (4)$$

- (a) Show that the horizontal and vertical components of the initial velocity are

$$u_0 = \frac{5\sqrt{2}(\sqrt{3} + 1)}{2} \quad \text{and} \quad v_0 = \frac{5\sqrt{2}(\sqrt{3} - 1)}{2}.$$

- (b) Consider the horizontal component of motion.

- (i) Equation (3) is variable separable. Determine u in terms of u_0 and t .
 (ii) Hence determine x in terms of u_0 and t .

- (c) Consider the vertical component of motion.

- (i) Use the answer to part (b)(i) and equation (4) to show that

$$\frac{d}{dt}((10 + u_0 t)v) = -10(10 + u_0 t).$$

- (ii) Show that $v = -\frac{10(10 + u_0 t)}{2u_0} + \frac{10(v_0 + \frac{100}{2u_0})}{10 + u_0 t}$ and hence find y .

- (d) Set $v = 0$ and hence find the time taken to reach the maximum height, as well as the coordinates of that point. Give the answers correct to two decimal places.

2. [Low-angle Trajectory] A particle is projected from the origin with velocity $u_0\mathbf{i} + v_0\mathbf{j}$ with units of m/s. Resistance is proportional to the square of the speed at each point in the trajectory and the constant of proportionality is km where m is the mass of the particle. Assume that $u_0 \gg v_0$ so that the speed at each point in the trajectory may be approximated by $(u^2 + v^2)^{\frac{1}{2}} \doteq u$. Also assume that the acceleration due to gravity is a constant $g \text{ m/s}^2$. Finally assume that there are no other forces acting on the particle.

- (a) By considering the horizontal and vertical components of force, write down equations for $\dot{u}$ and $\dot{v}$.

- (b) Use the result $\dot{u} = u \frac{du}{dx}$ to find u as a function of x .

- (c) Hence find $x(t)$ and thus determine $u(t)$.

- (d) Use the answers to parts (a) and (c) to show that $\frac{d}{dt}((1 + ku_0 t)v) = -g(1 + ku_0 t)$.

- (e) Show that $v = -\frac{g(1 + ku_0 t)}{2ku_0} + \frac{v_0 + \frac{g}{2ku_0}}{1 + ku_0 t}$ and hence find $y(t)$.

- (f) Use $x(t)$ and $y(t)$ to determine the Cartesian equation of the trajectory.

- (g) Determine $\frac{dy}{dx}$.

- (h) Hence show that the trajectory has a turning point at

$$\left(\frac{1}{2k} \log \left(\frac{2ku_0 v_0 + g}{g} \right), \frac{2ku_0 v_0 + g}{(2ku_0)^2} \log \left(\frac{2ku_0 v_0 + g}{g} \right) - \frac{2ku_0 v_0}{(2ku_0)^2} \right).$$

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3. [High-angle Trajectory] A particle is projected from the origin with speed 10 m/s at an angle of inclination of 75° . Resistance is proportional to the square of the speed at each point in the trajectory and the constant of proportionality is $\frac{1}{10}m$ where m is the mass of the particle. Assume that the acceleration due to gravity is 10 m/s^2 . Also assume that the speed may be approximated by $(u^2 + v^2)^{\frac{1}{2}} \approx |v|$. The equations of motion are thus

$$\dot{u} = -\frac{1}{10}u|v| \quad (5)$$

$$\text{and } \dot{v} = -10 - \frac{1}{10}v|v|. \quad (6)$$

- (a) Show that the horizontal and vertical components of the initial velocity are

$$u_0 = \frac{5\sqrt{2}(\sqrt{3}-1)}{2} \quad \text{and} \quad v_0 = \frac{5\sqrt{2}(\sqrt{3}+1)}{2}.$$

- (b) The Upward Journey: In this case $|v| = v$.

(i) Show that $v = 10 \tan(\theta - t)$, where $\tan \theta = \frac{1}{4}\sqrt{2}(\sqrt{3}+1)$.

- (ii) Explain why $0 \leq t \leq \theta$.

(iii) Then show that $y = 10 \log \left(\frac{\cos(\theta - t)}{\cos \theta} \right)$.

(iv) Hence show that the maximum height is $y_{\max} = 10 \log \left(\frac{1}{2} \sqrt{6 + \sqrt{3}} \right)$.

(v) Use part (i) and equation (5) to show that $\frac{d}{dt}(u \cos(\theta - t)) = 0$.

(vi) Hence show that $u = u_0 \cos \theta \sec(\theta - t)$.

- (vii) Given that $\int \sec \phi d\phi = \log |\sec \phi + \tan \phi| + C$, use the substitution $\phi = \theta - t$ in part (vi) to find an expression for x in terms of u_0 , θ and ϕ .

- (viii) Hence show that the x -coordinate of the turning point is

$$\frac{5\sqrt{2}(\sqrt{3}-1)}{\sqrt{6+\sqrt{3}}} \log \left(\frac{1}{4} \left(2\sqrt{6+\sqrt{3}} + \sqrt{2}(\sqrt{3}+1) \right) \right).$$

- (c) The Downward Journey: In this case $|v| = -v$.

- (i) Put $\tau = t - \theta$ and hence find $\frac{du}{d\tau}$ and $\frac{dv}{d\tau}$ in terms of u and v . Then write down the corresponding initial conditions for $\tau = 0$ in terms of u_0 and θ .

(ii) Use partial fractions to show that $v = -\frac{10(e^\tau - e^{-\tau})}{e^\tau + e^{-\tau}}$.

(iii) Hence show that $y = 10 \log \left(\frac{2 \sec \theta}{e^\tau + e^{-\tau}} \right)$.

- (iv) Put $y = 0$ into part (iii) to find τ and hence find the total time of flight.

(v) Now use part (ii) and equation (5) to show that $\frac{d}{d\tau}((e^\tau + e^{-\tau})u) = 0$.

(vi) Hence show that $u = \frac{(2u_0 \cos \theta)e^{-\tau}}{1 + e^{-2\tau}}$.

- (vii) Show that $x = u_0 \cos \theta \log(\sec \theta + \tan \theta)$ when $\tau = 0$. Then use part (vi) with this initial condition to find x in terms of u_0 , θ and τ .

- (viii) Use the value of τ in part (iv) to determine the range of the projectile in terms of u_0 and θ , then approximate this to three decimal places.

4. [Mid-angle Trajectory] A particle is projected from the origin with speed 10 m/s at an angle of inclination of 45° . Resistance is proportional to the square of the speed at each point in the trajectory and the constant of proportionality is $\frac{1}{10}m$ where m is the mass of the particle. Assume that the acceleration due to gravity is 10 m/s^2 . Also assume that the speed may be approximated by $(u^2 + v^2)^{\frac{1}{2}} \approx \sqrt{2}u \approx \sqrt{2}|v|$. The equations of motion are thus

$$\dot{u} = -\frac{\sqrt{2}}{10}u^2 \quad (7)$$

$$\text{and } \dot{v} = -10 - \frac{\sqrt{2}}{10}|v|v, \quad (8)$$

- (a) Use the answers to either Question 1 or Question 2, appropriately modified, to write down $u(t)$ and $x(t)$.
- (b) Use equation (8) to determine $v(t)$ and hence $y(t)$ for the upward journey. For what values of t is this solution valid?
- (c) Hence find the coordinates of the turning point of the trajectory, first in exact form, and then approximated to two decimal places.
- (d) Derive the formulae for $y(\tau)$ and $v(\tau)$ given in the text for the downward journey. Then replace g and k with the values specified here and simplify the results. In this case $\tau = t - \frac{\theta}{\sqrt{2}}$.
- (e) Use the formula for $y(\tau)$ to determine the total time of flight.
- (f) Hence determine the range of the trajectory correct to two decimal places.

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1(b)(i) $u = \frac{10u_0}{10 + u_0 t}$ (ii) $x = 10 \log \left(\frac{10 + u_0 t}{10} \right)$

(c)(iii) $y = \frac{10(100 - (10 + u_0 t)^2)}{(2u_0)^2} + \left(v_0 + \frac{100}{2u_0} \right) \frac{10}{u_0} \log \left(\frac{10 + u_0 t}{10} \right)$

Note that v simplifies to

$$v = \frac{10v_0 - 100t - 5u_0 t^2}{10 + u_0 t}$$

but this is harder to integrate.

(d) $t = \frac{1}{u_0}(\sqrt{2u_0 v_0 + 100} - 10) \approx 0.23 \text{ s}$

$x = 5 \log \left(\frac{2u_0 v_0 + 100}{100} \right) \approx 2.03 \text{ m}$

$y = \frac{10}{(2u_0)^2}((2u_0 v_0 + 100) \log \left(\frac{2u_0 v_0 + 100}{100} \right) - 2u_0 v_0) \approx 0.29 \text{ m}$

2(a) $\dot{u} = -ku^2$ and $\dot{v} = -g - kuv$. (b) $v = u_0 e^{-kx}$

(c) $x = \frac{1}{k} \log(1 + ku_0 t)$, $u = \frac{u_0}{1 + ku_0 t}$

(e) $y = \frac{g(1 - (1 + ku_0 t)^2)}{(2ku_0)^2} + \left(v_0 + \frac{g}{2ku_0} \right) \frac{1}{ku_0} \log(1 + ku_0 t)$

(f) $y = \frac{g(1 - e^{2kx})}{(2ku_0)^2} + \left(v_0 + \frac{g}{2ku_0} \right) \frac{x}{u_0}$

(g) $\frac{dy}{dx} = \frac{-ge^{2kx}}{2ku_0^2} + \left(v_0 + \frac{g}{2ku_0} \right) \frac{1}{u_0}$

3(b)(iii) At the top of the trajectory, $v = 0$ when $t = \theta$. (iv) $x = u_0 \cos \theta \log \left(\frac{\sec \theta + \tan \theta}{\sec \phi + \tan \phi} \right)$

(c)(i) $\frac{du}{d\tau} = \frac{uv}{10}$ and $\frac{dv}{d\tau} = -\frac{100 - v^2}{10}$

with $u = u_0 \cos \theta$ and $v = 0$ at $\tau = 0$.

(iv) $\tau = \log(\sec \theta + \tan \theta)$,

$t = \theta + \log(\sec \theta + \tan \theta)$

$= \tan^{-1} \left(\frac{\sqrt{2}(\sqrt{3}+1)}{4} \right)$

$+ \log \left(\frac{1}{4} \left(2\sqrt{6+\sqrt{3}} + \sqrt{2}(\sqrt{3}+1) \right) \right) \approx 1.625 \text{ s}$

(vi) $x = u_0 \cos \theta (\log(\sec \theta + \tan \theta))$

$+ \frac{\pi}{2} - 2 \tan^{-1}(e^{-\tau})$

(viii) $x_{\max} = u_0 \cos \theta (\log(\sec \theta + \tan \theta))$

$+ \frac{\pi}{2} - 2 \tan^{-1} \left(\frac{1}{\sec \theta + \tan \theta} \right) \approx 3.025 \text{ m}$

4(a) $u = \frac{5\sqrt{2}}{1+t}$, $x = 5\sqrt{2} \log(1+t)$

(b) $v = \sqrt{50\sqrt{2}} \tan(\theta - \sqrt{2}t)$ with $\theta = \tan^{-1} \sqrt{\frac{\sqrt{2}}{2}}$

and $y = 5\sqrt{2} \log \left(\frac{\cos(\theta - \sqrt{2}t)}{\cos \theta} \right)$ for $0 \leq t \leq \frac{\theta}{\sqrt{2}}$

(c) $y_{\max} = 5\sqrt{2} \log(\sec \theta) = 5\sqrt{2} \log \sqrt{1 + \frac{1}{2}\sqrt{2}}$

$\approx 1.89 \text{ m}$ at $x = 5\sqrt{2} \log \left(1 + \frac{\theta}{\sqrt{2}} \right) \approx 3.27 \text{ m}$

(g) $v = -5\sqrt{2\sqrt{2}} \times \frac{e^{\tau\sqrt{2}} - e^{-\tau\sqrt{2}}}{e^{\tau\sqrt{2}} + e^{-\tau\sqrt{2}}}$

$y = 5\sqrt{2} \log \left(\frac{2\sqrt{1 + \frac{1}{2}\sqrt{2}}}{e^{\tau\sqrt{2}} + e^{-\tau\sqrt{2}}} \right)$

$\log \left(\frac{\sqrt{2 + \sqrt{2}} + \sqrt{2}}{\sqrt{2}} \right) + \tan^{-1} \frac{\sqrt{2}}{2}$

(e) $t = \frac{\log \left(\frac{\sqrt{2 + \sqrt{2}} + \sqrt{2}}{\sqrt{2}} \right) + \tan^{-1} \frac{\sqrt{2}}{2}}{\sqrt{2}}$

(f) 5.67 m